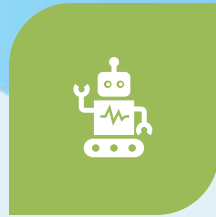


Supervised Machine Learning & Deep Learning with Python

Agenda



- What is supervised learning?



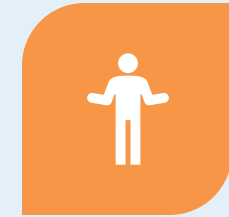
- ML algorithms & use cases



- Deep learning overview



- Implementation in python



- Evaluation & tuning

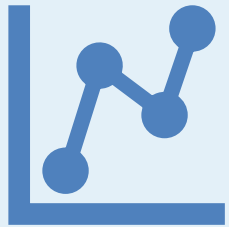


- Case study



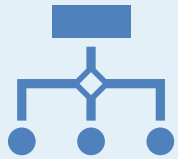
- Q&A

Introduction to Supervised Learning



- Supervised learning involves learning from labeled data to predict future outcomes.
- Examples: Spam filtering, sentiment analysis.

Supervised vs. Other Learning Types



Supervised: Uses labeled output



Unsupervised: No labels



Reinforcement: Learning via reward

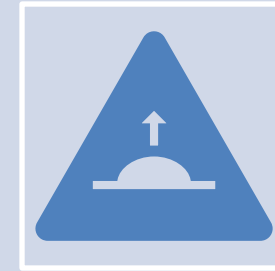


Supervised learning is foundational and widely applicable.

Task Types in Supervised Learning



- Regression: predict continuous values (e.g., Prices)



- Classification: Predict Categories (e.g., Spam Detection)

Classical ML Algorithms



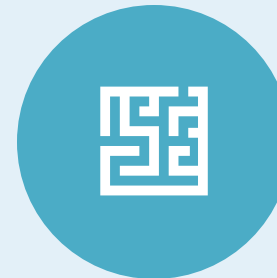
Linear/Logistic
Regression



Decision Trees &
Random Forests

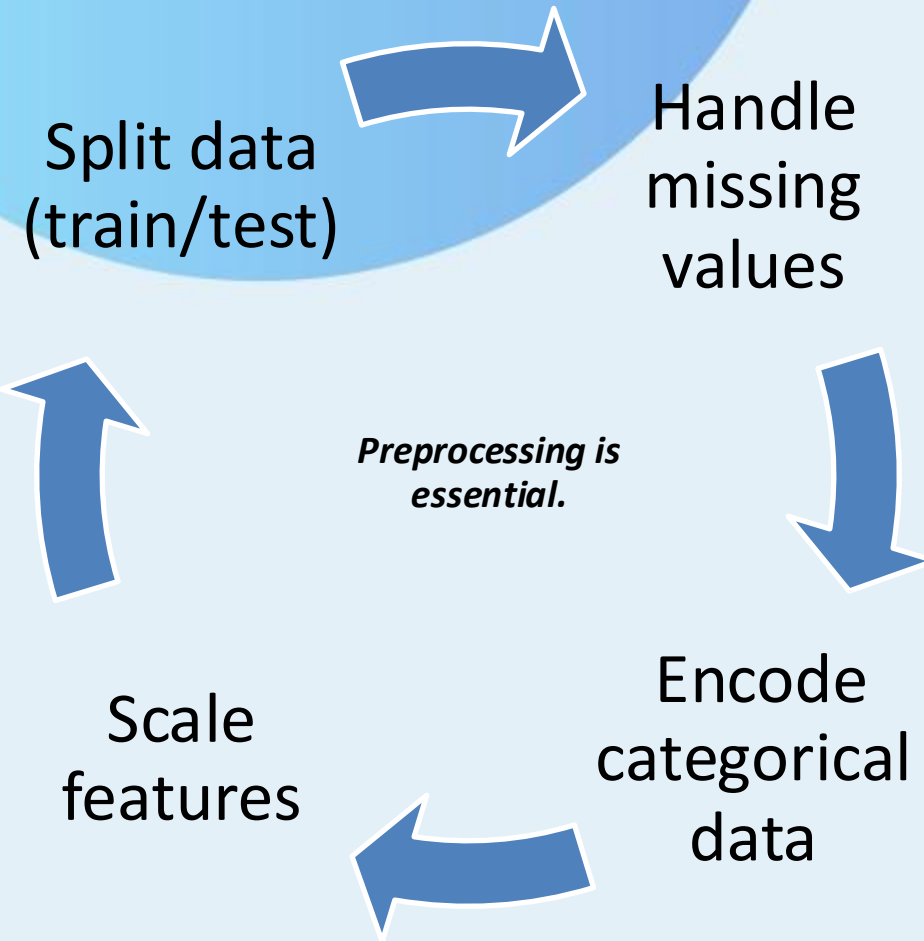


SVM, k-NN



Each algorithm has
specific problem
strengths.

Data Preprocessing



Feature Engineering

- Create interaction terms
- Add polynomial features
- Select important features

Model Evaluation

- Classification Model: Accuracy, Precision, Recall, F1, ROC-AUC
- Regression Model: MAE, MSE, RMSE, R^2

Use multiple metrics.

Machine Learning Pipeline

Data Collection and Exploration

Collect samples for training and evaluation

Identify ranges of data values

Compute histograms and plot data

Identify anomalies

Data Cleaning and Preprocessing

Fill missing values

Correct or remove errors

Normalize data

Partition datasets

Dimensionality reduction

Feature Engineering

Perform feature transformations

Select features

Model Training and Validation

Train models

Tune "hyper-parameters"

Evaluate on validation data

Model Evaluation and Selection

Evaluate models on test data

Select best performing model

Why Deep Learning?



Deep Learning Handles High-dimensional, Complex Data (Images, Speech, Etc.).

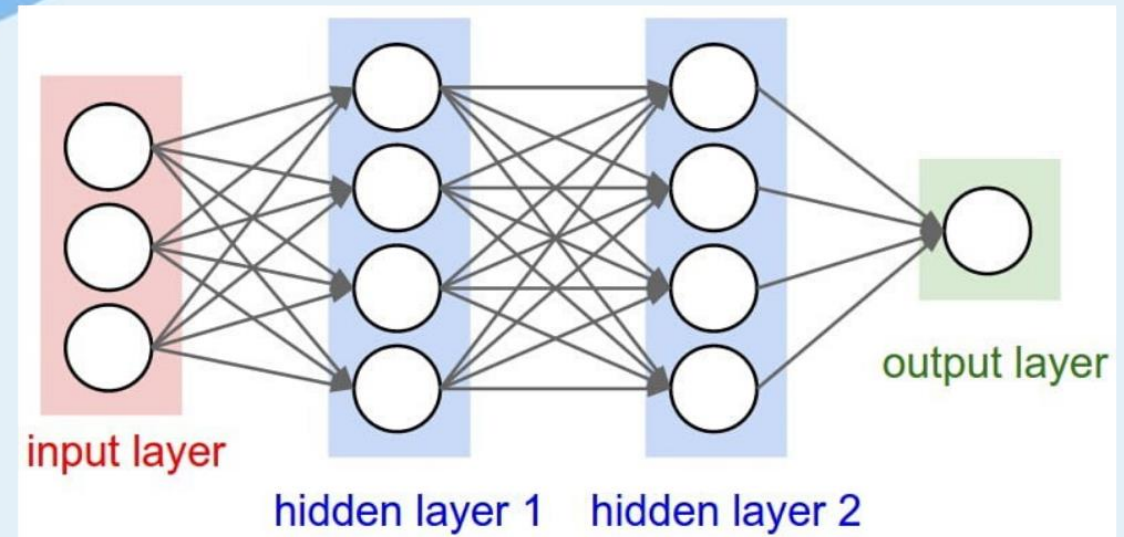


Goes Beyond What Classical ML Can Capture.

Neural Networks Basics

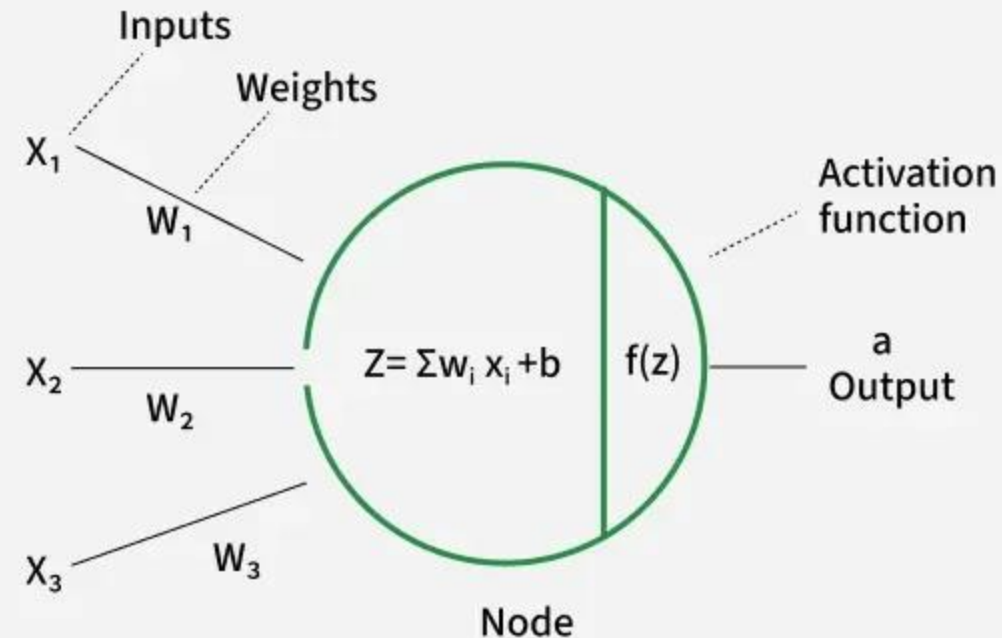
- Neural network components:
- Input layer
- Hidden layers
- Output layer

Activation functions: ReLU,
Sigmoid

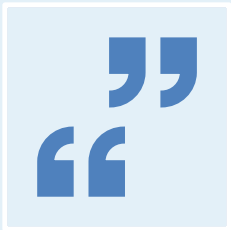


Neural Networks Basics

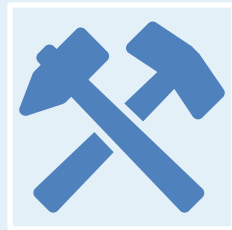
Activation functions in Neural Networks



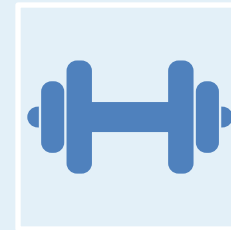
DL in Python



Popular libraries:
TensorFlow, Keras, PyTorch



Compile()



Fit()



Evaluate()

Overfitting in DL



- Dropout



- Early stopping



- Regularization



Monitor training vs. validation loss.

Model Explainability & Ethics



Tools: SHAP, LIME

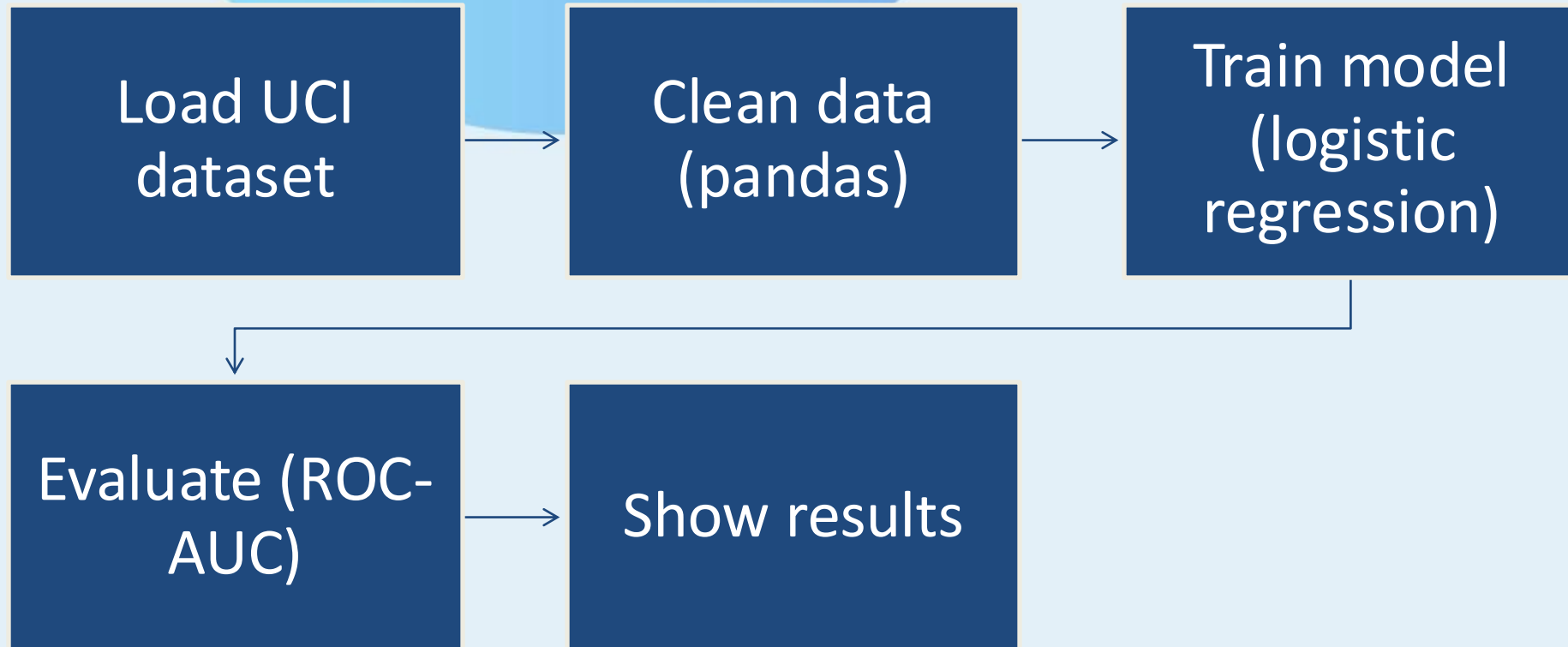


Topics: Fairness, bias, accountability



Promote responsible AI.

Case Study: Diabetes Prediction



Python Tools Recap



Data: Pandas,
Numpy



ML: scikit-learn



DL: keras,
tensorflow, torch



Visualization:
matplotlib, seaborn

Resources



Books: Hands-On ML, Deep Learning with Python



Courses: Coursera, fast.ai



Communities: Kaggle, GitHub

Final Thoughts

ML: Simple, interpretable

DL: Complex, powerful

Build, test, iterate.

Q&A



Questions?



Thank you!



We admonish you **to live**
for God and **connect with**
our community!



Connect with Us!

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